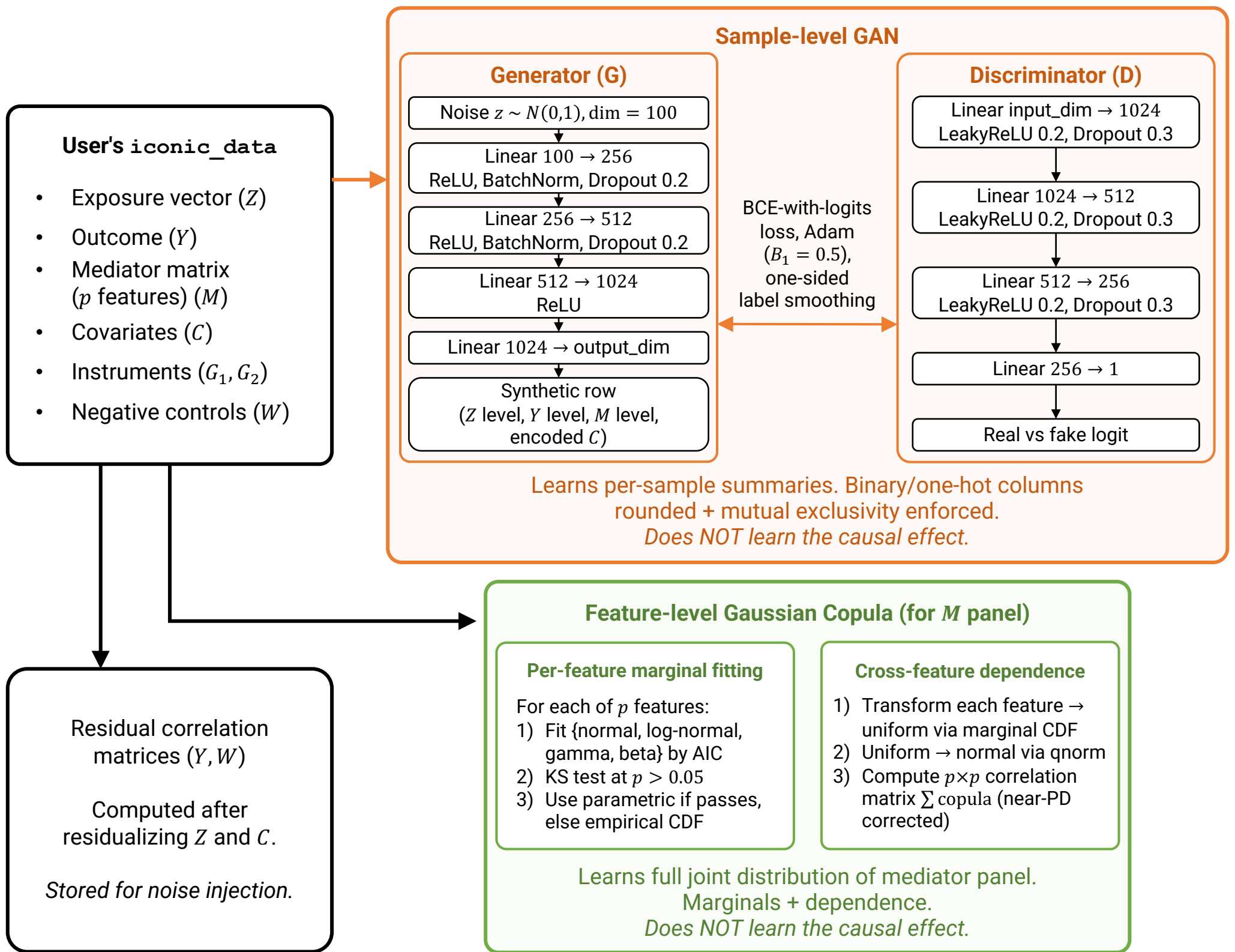


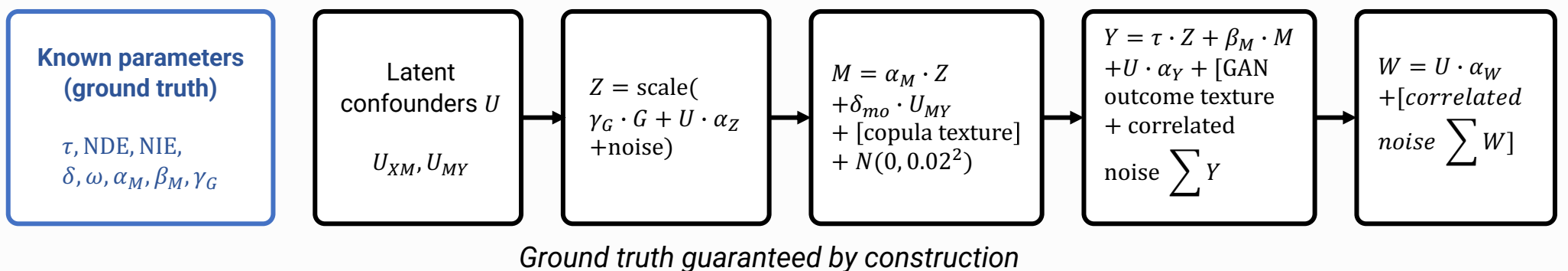
Data-Calibrated Simulation Mode; Generative Texture Pipeline

1) Texture Learning (trained on user's data)



Texture is drawn independently of U and never fed back into both Z and Y
(cannot open a hidden backdoor)

2) Structural Skeleton (imposed)



3) Synthetic Dataset

Synthetic (Z, M, Y, C, W, G)
with known ground truth + realistic texture

GAN learns nuisance texture for Z, Y, C . Copula learns nuisance texture for M . Causal effect ($\tau, \text{NDE}, \text{NIE}$) is NEVER learned, guaranteeing a closed form by imposed structural equations. Used for data-calibrated sensitivity and prospective analyses.